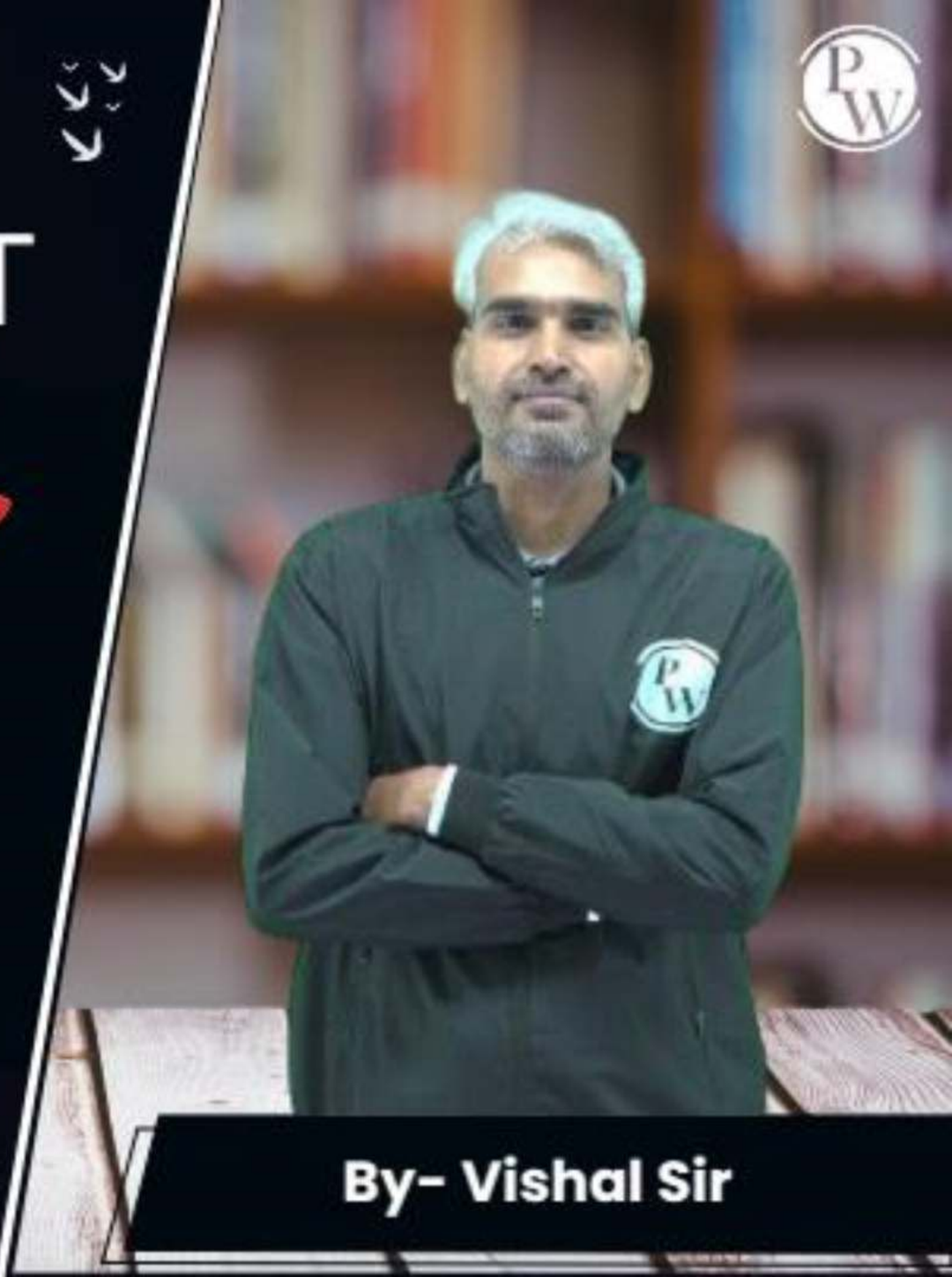
Computer Science & IT



Discrete Mathematics

Set Theory & Algebra

Lecture No. 23



By- Vishal Sir

Recap of Previous Lecture



Topic Groups w.r.t. Addition modulo ‘m’

Topic Groups w.r.t. Multiplication modulo ‘m’

Slide

Topics to be Covered



- **Topic** Order of an element in a group
- **Topic** Sub-group
- **Topic** Cyclic group

Slide



Topic : Order of an element in a group (G,*)

- Let $(G,*)$ be a finite group such that order of group $(G,*)$ is $|G|$.
- For an element $a \in G$, Order of element 'a' can be denoted by $O(a)$, and it is defined as the least positive integer 'n' such that

i.e. it can not be '0'

$$(a)^n = e \text{ (identity)}$$

$$= \underbrace{a * a * a * \ldots * a}_{n \text{ times } 'a'}$$

binary op^n w.r.t. group $(G,*)$

Note: Let $(G, *)$ is finite group and 'e' is the identity element of group $(G, *)$.

$$(e)^1 = e \text{ (identity)}$$

$$\therefore \boxed{O(e) = 1} \text{ Always}$$

eg. We know $\{1, -1\}$ is a group of order = 2 w.r.t. 'Multiplication'
Find order of every element of group. $O(G) = 2$

1 = identity $\therefore$ $\boxed{O(1) = 1}$

$(-1)^1 = -1$

$(-1)^2 = (-1)\cdot(-1) = 1$ (identity) $\therefore$ $\boxed{O(-1) = 2}$

eg: $\{1, \omega, \omega^2\}$ is a group w.r.t. multiplication

Find order of every element of group. $O(G) = 3$

1 = identity $\therefore$ $O(1) = 1$

$(\omega)^1 = \omega$

$(\omega)^2 = \omega^2$

$(\omega)^3 = \omega^3 = 1$ (identity)

$\therefore$ $O(\omega) = 3$

$(\omega^2)^1 = \omega^2$

$(\omega^2)^2 = \omega$

$(\omega^2)^3 = \omega^3 \cdot \omega^3 = 1 \cdot 1 = 1$ (identity)

$\therefore$ $O(\omega^2) = 3$

eg $\{1, -1, i, -i\}$ is a group w.r.t multiplication

Find order of Every element of group. $O(G) = 4$

1 = identity $\therefore$ $\boxed{O(1) = 1}$

$(-1)^1 = -1$

$(-1)^2 = 1$ (identity) $\therefore$ $\boxed{O(-1) = 2}$

$(i)^1 = i$

$(i)^2 = -1$

$(i)^3 = i^2 \cdot i = -1 \cdot i = -i$

$(i)^4 = i^2 \cdot i^2 = -1 \cdot -1 = 1$ (identity)

$\therefore$ $\boxed{O(i) = 4}$

$(-i)^1 = -i$

$(-i)^2 = i^2 = -1$

$(-i)^3 = -\{i^2 \cdot i\} = -\{-1 \cdot i\} = +i$

$(-i)^4 = i^2 \cdot i^2 = 1$ (identity) $\therefore$ $\boxed{O(-i) = 4}$



Topic : Properties w.r.t. Finite Group

① Order of identity element is always '1'

② Let $(G,*)$ be a finite group, then order of every element of group $(G,*)$ will be less than or equal to the order of group $(G,*)$, and order of every element of group $(G,*)$ divides the order of group $(G,*)$

It is Lagrange's Theorem

Let $(G,*)$ be a group of order $|G|$
then for all $a \in G$

$O(a)$ divides $|G|$



Topic : Properties w.r.t. Finite Group

③ Order of an element $a \in$ group $(G, *)$ is Exactly same as order of its inverse

i.e., $\boxed{O(a) = O(a^{-1})}$

④ Order of an element 'a' of group $(G, *)$ Can be '1' if and only if 'a' is the identity element w.r.t. group $(G, *)$

⑤ For an element 'a' of group $(G, *)$

if $O(a) = 2$ (i.e. $a * a = e$ (identity)), then element 'a' is inverse of itself

i.e. if $O(a) = 2$, then $a^{-1} = a$

* For any non-identity element $a \in$ Group

if $inv(a) = a$

then $O(a) = 2$

Q:. We know $\{0, 1, 2, 3, 4\}$ is a group w.r.t. $\oplus_5$.
Find the order of every element of the group.

0 = identity $\therefore$ $\boxed{O(0) = 1}$

$(1)^1 = 1$

$(1)^2 = 1 \oplus_5 1 = 2$

$(1)^3 = 1 \oplus_5 1 \oplus_5 1$

$= (1 \oplus_5 1) \oplus_5 1 = (1)^2 \oplus_5 1$

$= 2 \oplus_5 1 = 3$

$(1)^4 = (1)^3 \oplus_5 1 = 3 \oplus_5 1 = 4$

$(1)^5 = (1)^4 \oplus_5 1 = 4 \oplus_5 1 = 0$ (identity)

$\therefore$ $\boxed{O(1) = 5}$

$4 \oplus_5 1 = 0$ (identity) $\rightarrow$ $inv(1) = 4$

$\therefore$ $\boxed{O(1) = O(4) = 5}$

$(2)^1 = 2$

$(2)^2 = 2 \oplus_5 2 = 4$

$(2)^3 = (2)^2 \oplus_5 2 = 1$

$(2)^4 = (2)^3 \oplus_5 2 = 3$

$(2)^5 = (2)^4 \oplus_5 2 = 0$ (identity) $\therefore$ $\boxed{O(2) = 5}$

$= 3 \oplus_5 2 = 0$

$\rightarrow inv(2) = 3$

$\therefore$ $\boxed{O(3) = O(2) = 5}$

Q:. We know $\{0, 1, 2, 3, 4\}$ is a group w.r.t. $\oplus_5$.
Find the order of every element of the group.

Order of group = 5 ← it is a prime no.

We know order of every element must divide order of group

∴ Order of every element must divide '5' (Prime No.)

'5' can be divided by '1' or '5'

∴ Order (element) = 1 or Order (element) = 5

Possible only for identity

for Every non-identity element

∴ $O(0) = 1$

∴ $O(1) = O(2) = O(3) = O(4) = 5$

Note:- If order of group is a "prime number", then

Order (identity element) = 1

Order (Every non-identity element) = Order of the group = That prime no.

Sub-group

Sub-group

Let $(G, *)$ be a group, a non-empty subset H of set G is called a subgroup of group $(G, *)$ only if $(H, *)$ is a group.

(same binary opn as group $(G, *)$)

Sub-group

Note:- ① Every set is a subset of itself, so Every group is a subgroup of itself.

② If 'e' is the identity element of group $(G, *)$ then $(\{e\}, *)$ is always a subgroup of group $(G, *)$

$\{e\}$: Subset of G

* If $(G,*)$ is group such that 'e' is the identity element, then

group $(G,*)$ itself and group $(\{e\},*)$ are always sub-group of group $(G,*)$ and they are called "trivial subgroup" of group $(G,*)$

Any other sub-group of group $(G,*)$ is called Proper-sub-group of group $(G,*)$

eg. We know $\{1, -1, i, -i\}$ is a group w.r.t. multiplication

① $\{1, -1, i, -i\}$ is a trivial sub-group of above group

② $\{1\}$ is also a trivial sub-group of above group

③ $\{1, -1\}$ is a proper sub-group of above group

Note:- No other sub-group is exist w.r.t above group

eg:

We know $\{1, 3, 5, 7\}$ is a group of order $= 4$ w.r.t. $\otimes_8$

Following are the proper sub-groups of above group

1. $\{1, 3\}$ is a group of order $= 2$ w.r.t. $\otimes_8$
2. $\{1, 5\}$ is a group of order $= 2$ w.r.t. $\otimes_8$
3. $\{1, 7\}$ is a group of order $= 2$ w.r.t. $\otimes_8$

Properties w.r.t. Subgroup

(1) Let $(G,*)$ is a group, and H is any non-empty subset of set G, then

$(H,*)$ is a subgroup of group $(G,*)$

iff (1) H is closed w.r.t $*$

& (2) Identity w.r.t '$*$' exist in set H

& (3) inverse of every element of set H w.r.t. binary opn '$*$' exist in set H itself,

& (4) '$*$' is associative

Properties w.r.t. Subgroup

(1) Let $(G,*)$ is a group, and H is any non-empty subset of set G, then $(H,*)$ is a subgroup of group $(G,*)$ if and only if, $a * b^{-1} \in H$, $\forall a, b \in H$

Properties w.r.t. Subgroup

② Let $(G, *)$ be a group and $(H, *)$ be a subgroup of group $(G, *)$ such that $|G|$ = order of group $(G, *)$ and $|H|$ = order of subgroup $(H, *)$ then $|H|$ divides $|G|$

It is also Lagrange's Theorem

i.e.; Order of sub-group will always divide the order of original group.

Q:. Let $(G, *)$ be a group of prime order $= p$,
How many subgroup exists for group $(G, *)$

order of group $=$ p (Prime no.)

Order of sub-group always divides the order of group.

∴ Possible order w.r.t subgroup $=$ 1 or p

1: it will the sub-group w.r.t $\{e\}$

p: it will the subgroup w.r.t. original group $(G, *)$ itself.

Exactly '2' sub-groups & both of them will be trivial

Q:- Let $(G, *)$ be a group, and $(H_1, *)$ and $(H_2, *)$ are two subgroups of group $(G, *)$, then which of the following statements is/are true

S_1: $(H_1 \cup H_2, *)$ is also a sub-group of $(G, *)$

S_2: $(H_1 \cap H_2, *)$ is also a sub-group of $(G, *)$

Counter Example w.r.t. Statement S_1 :

$S = \{0, \pm1, \pm2, \pm3, \ldots\}$ (set of all integers) is a group w.r.t. addition

- Let $H_1 = \{0, \pm2, \pm4, \pm6, \pm8, \ldots\}$ all multiples of 2

& $H_2 = \{0, \pm3, \pm6, \pm9, \pm12, \ldots\}$ all multiples of 3

We can observe both H_1 & H_2 are group w.r.t addition.

And H_1 & H_2 are subset of set S.

∴ H_1 and H_2 are subgroup of above group

$H_1 \cup H_2 = \{0, \pm2, \pm3, \pm4, \pm6, \pm8, \pm9, \ldots\}$

$2 + 3 = 5 \notin H_1 \cup H_2$ ∴ Not closed w.r.t. addition

Hence $H_1 \cup H_2$ is not a group w.r.t. addition

and Hence $H_1 \cup H_2$ is not a subgroup of above group.

eg: $\{1, 3, 5, 7\}$ is a group w.r.t. $\otimes_8$

let $H_1 = \{1, 3\}$ is a sub group of above group

& $H_2 = \{1, 5\}$ is also a sub group of above group

$H_1 \cup H_2 = \{1, 3, 5\}$ it is not closed w.r.t. $\otimes_8$

$3 \otimes_8 5 = 7 \notin$ Set

Hence $H_1 \cup H_2$ is not a group w.r.t. $\otimes_8$

i.e. $H_1 \cup H_2$ is not a sub-group

Proof for Statement : S_2

Let $a, b \in H_1 \cap H_2$

then

$a, b \in H_1$

We know H_1 is a group

$\therefore a * b^{-1} \in H_1$

and

$a, b \in H_2$

We know H_2 is a group

$\therefore a * b^{-1} \in H_2$

$\therefore a * b^{-1} \in H_1 \cap H_2$

if $a, b \in H_1 \cap H_2$ then $a * b^{-1} \in H_1 \cap H_2$

$\therefore H_1 \cap H_2$ is definitely a group w.r.t. $*$



Topic : Properties w.r.t. Subgroup

③ Let $(G,*)$ be a group, and $(H_1,*)$ and $(H_2,*)$ be two subgroups of group $(G,*)$ then

$(H_1 \cup H_2, *)$ is a subgroup of group $(G,*)$ if and only if $H_1 \subseteq H_2$ or $H_2 \subseteq H_1$

↓ ↓

i.e. $H_1 \cup H_2 = H_2$ or $H_1 \cup H_2 = H_1$

↳ Otherwise $H_1 \cup H_2$ will not be closed w.r.t. binary opⁿ '$*$'.



Topic : Properties w.r.t. Subgroup

(4) Let $(G, *)$ be a group, and $(H_1, *)$ and $(H_2, *)$ be two subgroups of group $(G, *)$, then $(H_1 \cap H_2, *)$ is also a subgroup of group $(G, *)$

Cyclic Group



Topic : Cyclic group

* Let $(G, *)$ is a group. If there exists any element $a \in G$ such that Every element of set G can be represented in the form $(a)^n$ for some positive integer 'n', then

$(a)^n$ = $\underbrace{a * a * \ldots * a}_{n \text{ time } a}$

'n' : Can not be '0'.

group $(G, *)$ is called a Cyclic group. and element 'a' is one of the generator of cyclic group $(G, *)$

Note: In a cyclic group, Number of generators may be more than '1' as well

Note: Let $(G, *)$ be the group and 'e' is the identity element of the group.

$(e)^1 = e$

$(e)^2 = e * e = e$

$(e)^3 = e^2 * e = e * e = e$

$\vdots$

Identity element can not generate any other element except itself.

$\therefore$ Identity element can never be a generator of the set containing any other element apart from itself

Q: $\{1, -1\}$ is a group w.r.t. multiplication

Order of group = 2

Here $\bullet$ 1 = identity $\therefore$ can not generate any other element except itself

- $(-1)^1 = -1$
- $(-1)^2 = 1 = e$

} all the element of the group can be generated by '-1'

$\Downarrow$

Order $(-1) = 2$

$\therefore$ $\{1, -1\}$ w.r.t. multiplication is a cyclic group and '-1' is the generator of the cyclic group.

① Check whether the group $\{1, \omega, \omega^2\}$ w.r.t. multiplication is a cyclic group or not. if cyclic then find all the generators of the cyclic group

$O(G) = 3$

1 = identity (Not a generator)

$O(1) = 1 \neq O(G)$

$(\omega)^1 = \omega$

$(\omega)^2 = \omega^2$

$(\omega)^3 = \omega^3 = 1$ = identity

$O(\omega) = 3 = O(G)$

All elements of set

$\therefore \omega$ is a generator

$(\omega^2)^1 = \omega^2$

$(\omega^2)^2 = \omega$

$(\omega^2)^3 = 1$ = identity

$O(\omega^2) = 3 = O(G)$

All elements of set

$\therefore \omega^2$ is a generator

$\{1, \omega, \omega^2\}$ w.r.t. multiplication is a cyclic group, and ω & ω^2 are the generators

(2) Check whether the group $\{1, -1, i, -i\}$ w.r.t. multiplication is a cyclic group or not. if cyclic then find all the generators of the cyclic group

$O(G) = 4$

1 = identity (Not a generator)

$O(1) = 1 \neq O(G)$

$(-1)^1 = -1$

$(-1)^2 = 1$ } = identity

$O(-1) = 2 \neq O(G)$ →

$(-1)^3 = -1$ } Repeat

$(-1)^4 = 1$ } = identity

$(-1)^5 = -1$ } Repeat

$(-1)^6 = 1$ } = identity

If order of an element is not Equal to the order of the group, then it will generate the identity element before it has generated all the elements of the set. And after that if we increas the Power then it will start repeating the elements that it has already generated and will not be able to generate the remaining elements so such an element can never be the generator.

(2) Check whether the group $\{1, -1, i, -i\}$ w.r.t. multiplication is a cyclic group or not. if cyclic then find all the generators of the cyclic group

$O(G) = 4$

1 :- identity (Not a generator)

$O(1) = 1$

$(-1)^1 = -1$

$(-1)^2 = 1$ = identity

$(-1)^3 = -1$

$(-1)^4 = 1$

$(-1)^5 = -1$

$(-1)^6 = 1$

elements will be repeated

$O(-1) = 2 \neq O(G)$

$(i)^1 = i$

$(i)^2 = i^2 = -1$

$(i)^3 = i^2 \cdot i = -i$

$(i)^4 = i^2 \cdot i^2 = 1$ = identity

$O(i) = 4 = O(G)$

All elements

$\therefore$ 'i' is a generator

$(-i)^1 = -i$

$(-i)^2 = i^2 = -1$

$(-i)^3 = +i$

$(-i)^4 = 1$ = identity

$O(-i) = 4 = O(G)$

All elements

$\therefore$ '$-i$' is also a generator

$\{1, -1, i, -i\}$ w.r.t. multiplication is a cyclic group & 'i' & '$-i$' are generators

H.W. ③ Check whether the group $\{1, 3, 5, 7\}$ w.r.t. $\otimes_8$ is a cyclic group or not. If cyclic then find all the generators of the cyclic group $(O(G)=4)$



2 mins Summary

Topic Order of an element in a group

Topic Sub-group

Topic Cyclic group

Slide

THANK - YOU